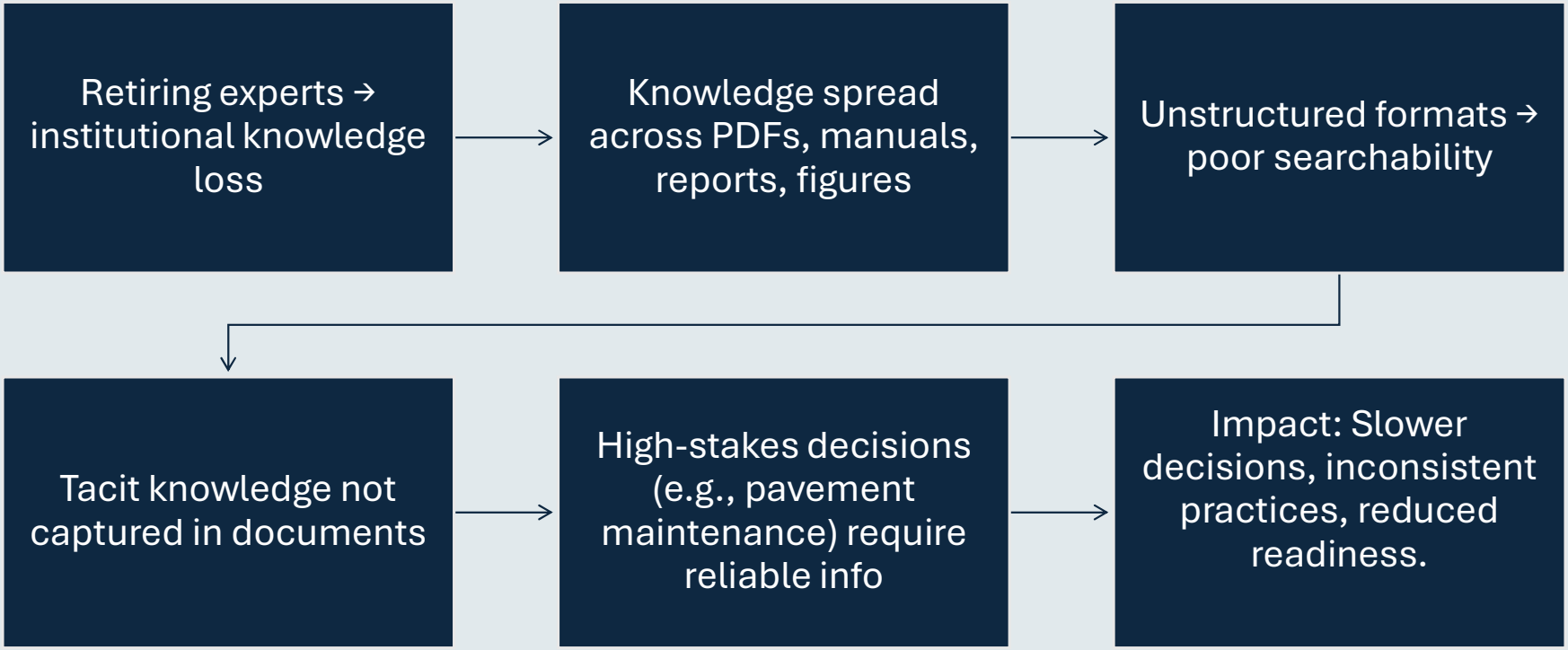
Developing an AI Assistant for Knowledge Management in State DOT

Tyler Blue
Andrew Moy

The Knowledge Management Problem



Proposed Solution: Multi-Agent RAG

- Four-agent architecture:
  - Retriever: Finds relevant text + figure descriptions
  - Generator: Produces grounded, structured responses
  - Evaluator: Checks correctness, clarity, relevance
  - Query Refiner: Iteratively improves weak queries

Making Figures Searchable

- DOT documents rely heavily on figures; needed way to include them in vector database
- Qwen3-VL-2B-Instruct converts figures into:
  - Captions
  - Semantic Summaries
  - Quantitative trends
  - Comparative insights

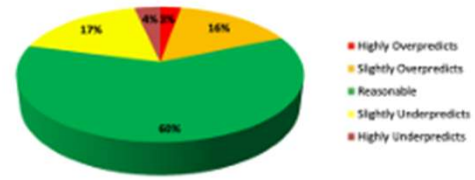


Figure 8. Opinions of TxDOT's Practitioners of Newly-Developed Pavement Performance Prediction Models.

(b) Pie chart adopted adapted from Gharaibeh et al. [58].

Qwen3-VL-2B-Instruct generated description: This figure is a pie chart that displays the opinions of TxDOT's practitioners regarding newly-developed pavement performance prediction models. The chart is divided into five segments, each representing a different level of prediction accuracy with corresponding percentages. The largest segment, colored green, accounts for 60% and represents "Reasonable" predictions. The second-largest segment, yellow, represents "Slightly Underpredicts" at 17%. The smallest segments are "Highly Overpredicts" (red, 4%) and "Slightly Overpredicts" (orange, 16%). The legend on the right side of the chart identifies each category by its color and label.

Query Refinement

Evaluator checks response for clarity, correctness, and completeness

If evidence is missing → triggers refinement loop

Query Refiner rewrites query for specificity, clarity, and relevance

Refined query gets sent back to Retriever and can loop up to k times

Result: More complete retrieval and better answers

Case Study

- Ingested 500 DOT documents
- Extracted text + figures
- Embedded into vector database
- Evaluated on 100 domain-specific questions
- Results:
  - Single-Pass RAG: 58% Recall@5
  - Multi-query RAG: 94% Recall@3

Conclusion

- Knowledge fragmentation and expert loss are not unique to DOTs
- Many organizations face similar challenges with unstructured, siloed information
- Traditional solutions (manual training, keyword search) do not scale effectively
- Takeaway: AI-driven knowledge systems can preserve expertise and improve decision-making at scale

Sources

Amaram et al., 2026

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arXiv:2603.03302